

ESE5180 Lab 1

Wireless Communications

Team 15

Team member	Email
Lei, Si Wei	laialex@engineering.upenn.edu
Yu, Alexander	ayu2126@engineering.upenn.edu

Repository: github.com/Alex-lei-1/ese5180f26-lab1-t15

1.1 LoRa Range Challenge

Result	Value
Longest successful distance	478 m
RSSI at longest distance	-99 dBm
SNR at longest distance	-6 dB

The screenshot shows a VS Code editor with the file explorer on the left displaying a project structure for 't26-lab1-t12'. The main editor window shows the source code for 't1_1_receive.c'. The code includes headers, defines, and functions for configuring and receiving LoRa packets. The serial monitor window at the bottom shows a list of received packets with timestamps, RSSI, SNR, and payloads. The payloads are 'ese5180t12-3', 'ese5180t12-3', 'ese5180t15-2', 'ese5180t15-2', 'ese5180t05-7', 'ese5180t05-7', 'ese5180t15-3', 'ese5180t15-3', 'ese5180t12-4', and 'ese5180t12-4'.

Figure 1. Team 15 packets in the range-test serial capture.

Frequency	Power	BW	SF	CR	Preamble
433.92 MHz	-10 dBm	125 kHz	SF11	4/8	16

Why is this the case? Check out the firmware source—if you wanted to avoid interference from other student LoRa boards, what might you do?

Raw LoRa has no device identity at the physical layer, so a receiver accepts packets with matching radio settings. Our receiver checks the `ese5180t15-` prefix and counter and ignores other teams' packets.

What firmware and hardware configurations helped increase the successful transmission range?

SF11, BW125, CR4/8, a 16-symbol preamble, line of sight, correct antenna orientation, and keeping the antenna away from metal improved range.

What are some disadvantages you've noted when maximizing distance this way?

The configuration has a low data rate, long airtime, higher energy per packet, and more channel occupancy.

What configurations are realistic in a real-world product?

Use the shortest airtime and lowest transmit power that still provide the required range and link margin, while obeying regional radio rules.

1.2 Trading Range for Bandwidth

Parameter	1.1 range	1.2 bandwidth
Bandwidth	125 kHz	500 kHz
Spreading factor	SF11	SF5
Coding rate	4/8	4/5
Preamble	16 symbols	12 symbols
12-byte airtime	857 ms	4 ms
Nominal coded bit rate	0.336 kbps	62.5 kbps

The fastest measured packet airtime was **4 ms**. Frequency remained 433.92 MHz and TX power remained -10 dBm. Increasing bandwidth to 500 kHz, reducing the spreading factor to SF5, using CR4/5, and using the minimum valid 12-symbol SF5 preamble produced this result.

```
RX RSSI: -54 dBm | SNR: 9 dB | Payload: ese5180t15-1
RX RSSI: -52 dBm | SNR: 9 dB | Payload: ese5180t15-2
```

What is the “Spreading Factor”? What transmission quantities does the spreading factor affect? What is the tradeoff?

The spreading factor determines the number of chirps per symbol (2^{SF}). A higher SF increases symbol duration, receiver sensitivity, range, airtime, and packet energy while reducing data rate. A lower SF does the reverse.

What kinds of applications would suit the low-bandwidth option? Name at least two. Why would the tradeoff be worth it?

Environmental or agricultural sensors, utility meters, and remote asset trackers send small, infrequent messages, so longer range and battery life are more valuable than high throughput.

1.3 Energy Check

Profile	TX power	Peak TX current
Low	-9 dBm	39.20 mA
High	+22 dBm	157.89 mA

Both builds used 915 MHz, BW125, SF10, CR4/5, an 8-symbol preamble, a 12-byte payload, and a 2-second interval. Only TX power changed.

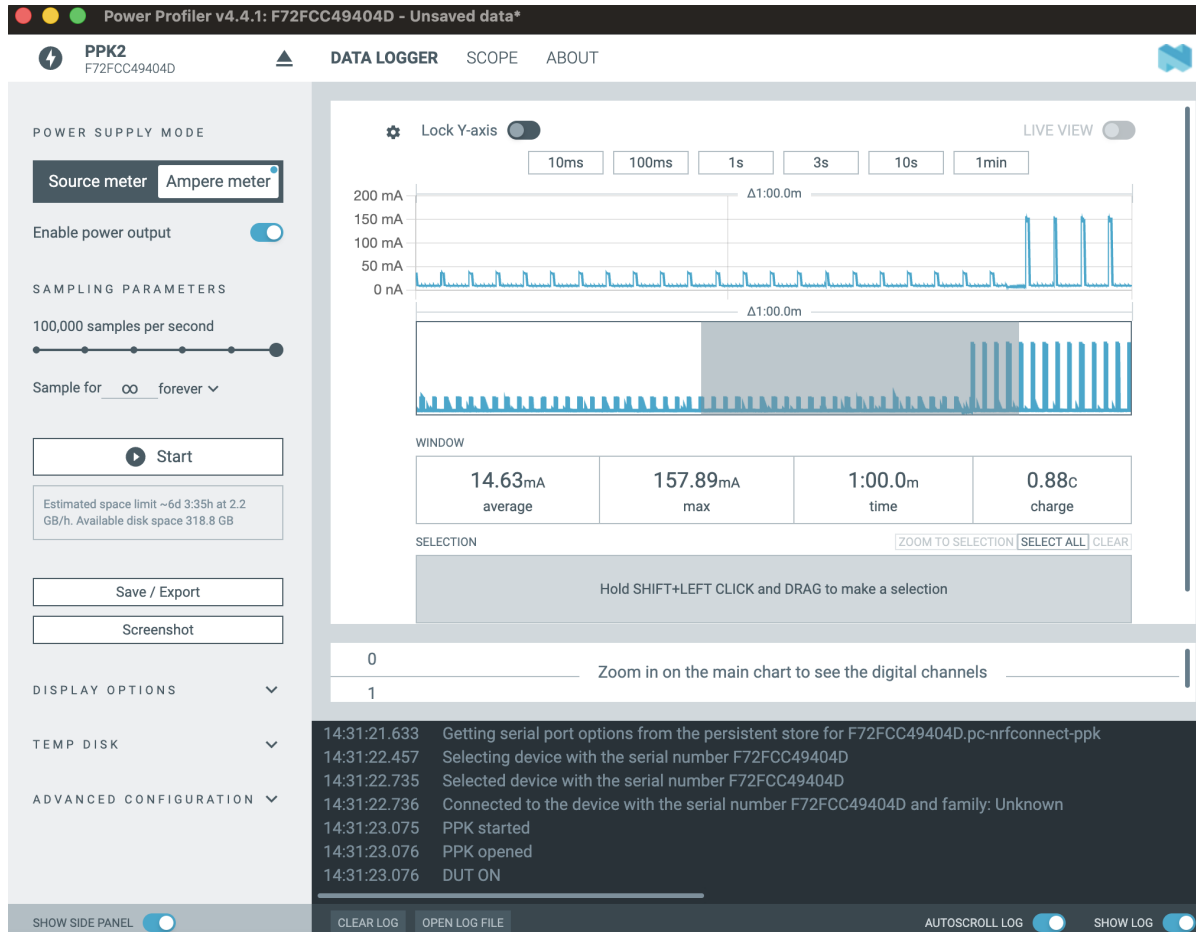


Figure 2. One continuous PPK2 recording showing the low-to-high TX-current transition.

Beyond just transmit power, what else will affect the current draw and battery life of a LoRa device? Could a coin cell battery power the largest Tx current draw measured? AA batteries? LiPo?

Airtime, message frequency, retries, receive windows, MCU sleep current, peripherals, regulator efficiency, and self-discharge affect battery life. A coin cell is generally unsuitable for the measured 157.89 mA pulse because of voltage sag. AA cells or a suitable LiPo can supply it if their datasheet current limits are met.

2. LoRaWAN

2.1 Modified LoRaWAN code

Files: 2_lorawan/src/main.c, 2_lorawan/prj.conf, and 2_lorawan/uplink_decoder.js

```
{"name": "Alex", "team": "15", "board": "WL55JC"}
```

2.2 Latest decoded payload, live data, and serial monitor

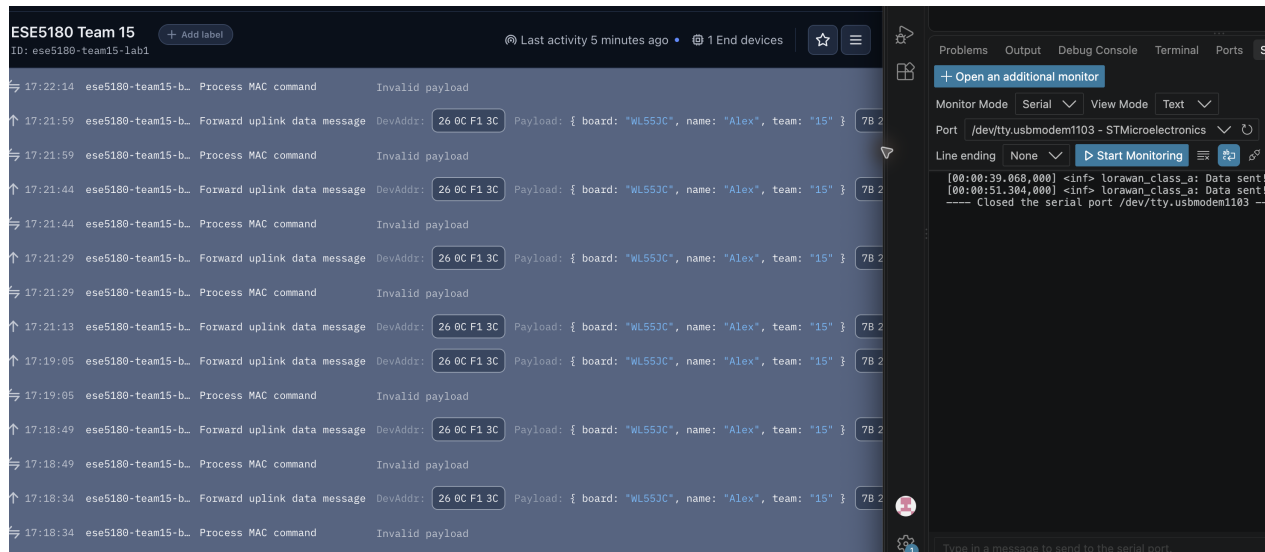


Figure 3. TTN decoded Team 15 JSON while the serial monitor reported successful sends.